

ANNA BURGESS

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EDUCATION

Harvard College | Cambridge, MA

Exp. May 2027

Bachelor of Science in Mechanical Engineering, Minor in Mathematical Sciences, GPA: 3.89

Relevant Coursework: Computer-Aided Machine Design, Mechanical Systems, Solid Mechanics, Computational Mechanics, Thermodynamics, Heat Transfer, Robotics

RESEARCH EXPERIENCE

Research Intern | Mechatronics and Haptic Interfaces Lab, Rice University | Houston, TX

May 2025 – July 2025

PI: Prof. Marcia K. O'Malley | **Research Mentor:** Dr. Shane King

- Designed customizable TPU housings integrating FSRs and electronics for a center-of-pressure sensing insole and force-measuring scoliosis brace
- Developed equation-driven, multi-material SolidWorks models and additive manufacturing processes for flexible TPU, conductive TPU, embedded electronics, and serviceable sensor architectures
- Designed and conducted FSR calibration and validation experiments using an Instron, completing 125 calibration trials with controlled loading to 200 N and 500-Hz data acquisition
- Validated 4 scoliosis-brace sensor prototypes through Instron testing, then fabricated 10 units for handoff to Baylor Scott & White for clinical testing
- Identified wire-management and conductive-material limitations affecting insole sensor consistency and iterated designs based on experimental results

Undergraduate Research Assistant | Bioengineering Ability Lab, Harvard University | Cambridge, MA

August 2025 – May 2026

PI: Prof. Patrick Slade | **Research Mentor:** Jordan Feldman

- Redesigning an existing visual-perturbation headset for balance research, replacing failure-prone acrylic and taped assemblies with lightweight 3D-printed PLA components
- Designed modular assembly using heat-set inserts, removable fabric/Velcro weight covers, and mechanically constrained mirrors to improve durability and serviceability
- Developed slotted, set-screw, and friction-fit interfaces to securely retain mirrors while enabling straightforward assembly and replacement
- Redesigning the servo-driven rotational axis with press-fit hex interfaces and screw-access features to simplify manufacturing and assembly
- Iterated CAD and 3D-printed prototypes to resolve structural and assembly issues; delivered completed headset to researchers for experimental testing

PRESENTATIONS

- **Burgess, A., King, S., Stovicek, K., & O'Malley, M. K.** "3D-Printed Thermoplastic Polyurethane Housings for Force-Sensing Resistors in Biomedical Research." Poster presented at the Rice Summer Undergraduate Research Symposium, Rice University, Houston, TX, August 2025.

ENGINEERING LEADERSHIP

Chief Engineer and Project Manager | Harvard Undergraduate Aerospace Collective – Rocketry | Cambridge, MA

Oct. 2025 – Present

- Founded Harvard's 10,000-ft IREC competition rocketry program and led 5 members through NAR Level 1 high-power rocketry certification
- Lead development of a liquid bipropellant regeneratively cooled rocket engine with an additively manufactured metal combustion chamber

- Coordinate four technical subteams through assembly, integration, and static-fire testing of a 250-lbf, 5,000-N·s bipropellant rocket
- Initiated a 3–5 km LoRa telemetry project and designed avionics housings for flight hardware tested aboard a micro-G liquid rocket
- Manage team budget, project schedule, transportation logistics, onboarding, outreach, media, and organizational operations
- Collaborated with 5 board members to form the Harvard Undergraduate Aerospace Collective, unifying resources across Rocketry, Satellite, and UAV teams
- Lead design of a modular rocket test stand with adjustable restraints, configurable mounting, and thrust redirection for varied test environments

Machining Lead | Harvard Undergraduate Robotics Club | Cambridge, MA

Oct. 2024 – Present

- Machined precision power-transmission components to ± 0.003 -in tolerances for a University Rover Challenge Mars rover
- Selected manufacturing processes and redesigned components to reduce machining complexity, setup time, and fabrication difficulty
- Co-led a 40-member mechanical team, allocating work across arm, chassis, suspension, and science modules

Aerostructures Lead | Harvard Undergraduate Rocket and Propulsion Group | Cambridge, MA

Oct. 2024 – Oct. 2025

- Performed SolidWorks FEA on rocket bulkheads under high loads to verify structural integrity
- Designed and fabricated an 800-lbf rocket engine static test stand with a $2\times$ factor of safety

INDUSTRY EXPERIENCE

Mechanical Engineering Intern | Flex | Austin, TX

May 2026 – Aug. 2026

- Developed 26 production tooling and fixture projects to reduce cycle time, prevent component damage, and improve assembly repeatability
- Designed a machined universal product nest for a 13-station automated assembly line to withstand 65°C processing without deformation
- Produced 178 SolidWorks parts/assemblies and 25+ peer-reviewed engineering drawings, applying dimensional tolerancing and tolerance stack-up analysis
- Designed screw-mask fixtures to protect components, indicate fastener locations, and communicate torque sequence without repeated reference to work instructions
- Developed handling and support fixtures to prevent PCB flexing, protect bottom-side components, and improve repeatability during assembly
- Designed alignment fixtures to address operator-reported quality and workflow issues, including a vacuum-actuated placement guide for difficult-to-reach product labeling
- Iterated designs through rapid prototyping and direct feedback from operators, engineers, and cross-functional engineering teams
- Managed up to 6 active projects simultaneously while balancing changing priorities, customer needs, and production constraints

Welding Intern | McCannic Shop | Evant, TX

June 2023 – Aug. 2023

- Fabricated a front-mounted vehicle-moving device from angle iron and tires to reposition vehicles while minimizing towing-related damage
- Cut, fit, and MIG-welded steel components during fabrication and repair work

TEACHING EXPERIENCE

Head Course Assistant | Introduction to Computer-Aided Machine Design, Harvard | Cambridge, MA

Oct. 2024 – Present

- Instruct 1–2 teams of 4–6 students each semester in machining and the design and fabrication of a competition robot
- Lead weekly office hours for groups of 10–20 students covering statics, G-code, SolidWorks, GD&T, and manufacturing processes
- Support classes of 40–60 students through technical instruction, troubleshooting, and individualized academic assistance

- Grade technical assignments and SolidWorks design work, providing detailed individualized feedback

Maker Fellow | REEF Makerspace, Harvard | Cambridge, MA

Aug. 2025 – May 2026

- Independently managed operations in a multidisciplinary creative makerspace
- Assisted students with laser cutting, 3D printing, soldering, desktop milling, sewing, embroidery, heat-press operation, and related fabrication processes
- Regularly checked in with student groups to provide individualized project support and troubleshooting
- Designed example projects and wrote documentation to expand use of an underutilized desktop milling machine

Private Tutor | Lampasas, TX

Summers 2023, 2024, 2025

- Provided individualized instruction from elementary through GED-level subjects, adapting lessons to student needs and learning styles
- Developed lesson plans and hands-on learning activities for an elementary student addressing COVID-era learning gaps; tracked progress and communicated with parents
- Prepared an adult learner for the GED through targeted instruction and practice across required subjects; student successfully earned his GED

HONORS & AWARDS

- 2nd Place, Egg-Drop Design Competition — Dynamics, Harvard | 2025
- 2nd Place, Bridge Design Competition — Statics, Harvard | 2025
- 2nd Place, Turf Wars Robot Design Competition — Computer-Aided Machine Design, Harvard | 2023
- Valedictorian, Lampasas High School | 2023
- Commended Student, National Merit Scholarship Program | 2023
- National Rural and Small Town Recognition Program | 2022
- AP Scholar with Distinction | 2022 & 2023

TECHNICAL SKILLS

Manufacturing: Manual/CNC Turning and Milling, MIG/TIG Welding, Waterjet Cutting, Plasma Cutting, Laser Cutting, 3D Printing, Soldering

CAD/CAE/CAM: SolidWorks, Fusion 360, Abaqus, COMSOL

Programming & Controls: Python, MATLAB, Arduino

CNC Programming: G-code, Conversational Programming (ProtoTRAK)

Engineering Methods: GD&T, Dimensional Tolerancing, Tolerance Stack-Up Analysis, FEA, Design for Manufacturability (DFM), Rapid Prototyping, Engineering Drawings

Certifications: Certified SOLIDWORKS Associate (CSWA), NAR Level 1 High Power Rocketry

ADDITIONAL EMPLOYMENT

Server/Fryer | Quincy Grille, Harvard | Cambridge, MA | Sept. 2023 – May 2026

Server/Bartender/Lane Server | Putters & Gutters | [Location] | July 2021 – Jan. 2025